
✓ **LIKELY EXAM QUESTIONS FROM YOUR LECTURE**

A. Long Theory Questions (8–15 marks)

These are highly predictable:

1. Explain the architecture of an Expert System with a neat diagram.

Must mention:

- Knowledge base
- Working memory
- Inference engine
- Communication module
- User interface

2. Describe the steps involved in the development of an Expert System.

Mention the 5 stages:

1. Identification
2. Conceptualization
3. Formalization
4. Implementation
5. Testing

3. What are the different components of the inference engine? Explain forward chaining, backward chaining, and abduction.

You must define:

- How each works
- Differences

- When each is used
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4. What is Knowledge Engineering? Explain knowledge acquisition bottleneck.

Talk about:

- Cognitive barrier
 - Linguistic barrier
 - Representation barrier
 - Model creation problem
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5. Explain Knowledge Representation Techniques used in Expert Systems.

Must include:

- Rules
 - Semantic nets
 - Frames
 - Scripts
 - Object-oriented representation
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B. Short Notes (3–6 marks)

These come very frequently:

- 1. Expert system vs conventional program**
- 2. Semantic nets**
- 3. Frames and inheritance**
- 4. Explanation facility (HOW and WHY answers)**
- 5. Reasoning under uncertainty**
- 6. CLIPS expert system tool**
- 7. RADEX expert system**

8. Christiansen ET method

9. A priori vs A posteriori knowledge

10. Working memory

C. Likely Short Answer Questions (1–3 marks)

1. Define an Expert System.

2. What is a shell? (e.g., EMYCIN)

3. What is a knowledge base?

4. What is inductive learning?

5. Give one example of an IF–THEN rule.

6. What is forward chaining?

7. What does the inference engine do?

8. State the Christiansen formula for ET.

D. Diagram-Based Questions

These may ask you to **draw and label**:

1. Architecture of an Expert System

2. Forward chaining flow diagram

3. Backward chaining flow diagram

4. Generalization tree (from induction section)

E. Application / Problem Solving Questions

1. Using given climate data, calculate ET using the Christiansen formula.

They may give:

- Ra
- CT

- CH
- CU
- CS
- CM
- CE

And ask you to compute ET.

2. Explain how RADEX uses rules and calculations in its expert system.

F. Very Likely “Compare / Differentiate” Questions

- 1. Forward chaining vs backward chaining**
 - 2. Expert system vs knowledge-based system**
 - 3. Declarative vs procedural knowledge**
 - 4. Conventional programs vs expert systems**
 - 5. A priori vs A posteriori knowledge**
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TOP FIVE QUESTIONS MOST LIKELY TO APPEAR

If you can answer these perfectly, you will pass comfortably:

- 1. Explain the architecture of an Expert System with a diagram.**
 - 2. Differentiate between forward chaining, backward chaining, and abduction.**
 - 3. Explain knowledge representation techniques.**
 - 4. What is knowledge engineering? Discuss the knowledge acquisition bottleneck.**
 - 5. Write short notes on CLIPS and RADEX.**
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1) How to compute ET with Christiansen formula

Formula:

$$\text{ET} = 0.473 \times R_a \times C_T \times C_H \times C_U \times C_S \times C_M \times C_E$$

Step-by-step method

1. Make sure all inputs are for the same time period (e.g., daily).
 - (R_a) = extra-terrestrial radiation (use units given in the question).
 - ($C_T, C_H, C_U, C_S, C_M, C_E$) = dimensionless coefficients for temperature, humidity, wind, sunshine, elevation, consumptive use.
2. Multiply the coefficients together first (or sequentially) to reduce mistakes.
3. Multiply the product by (R_a).
4. Multiply that by 0.473 → that gives ET.
5. State the unit (same time basis as (R_a); e.g., mm/day if coefficients and R_a are consistent with that).

Worked example (use in exam if numbers provided)

Assume (example values):

- ($R_a = 15.0$) (units consistent with ET, e.g., $\text{MJ m}^{-2} \text{ day}^{-1}$ or the exam's provided unit)
- ($C_T = 1.05$)
- ($C_H = 0.95$)
- ($C_U = 1.10$)
- ($C_S = 0.90$)
- ($C_M = 1.02$)
- ($C_E = 0.85$)

Compute step-by-step:

1. Multiply coefficients:

- $(1.05 \times 0.95 = 0.9975)$
- $(\times 1.10 = 1.09725)$
- $(\times 0.90 = 0.987525)$
- $(\times 1.02 = 1.0072755)$
- $(\times 0.85 = 0.856184175)$

2. Multiply by (R_a) : $(15.0 \times 0.856184175 = 12.842762625)$

3. Multiply by 0.473: $(0.473 \times 12.842762625 = 6.074626721625)$

Answer: $(\displaystyle \text{ET} \approx 6.07)$ (same unit as expected by the exam — state it, e.g., mm/day).

Exam tip: show the coefficient multiplication steps and final multiplication by 0.473 — examiners like clear work.

2) How RADEX uses rules and calculations (short, exam-style answer)

What RADEX is: an expert system for predicting evapotranspiration (ET) that combines **rule-based logic** with **numerical formulas** (like Christiansen).

How it works (flow):

1. **Inputs:** climate data — temperature, humidity, wind speed, sunshine hours, elevation, latitude/month (for (R_a)), etc.
2. **Preprocessing:** compute intermediate values (e.g., convert sunshine to solar radiation (R_s) from (R_a) and n/N if needed).
3. **Rule layer** (IF–THEN rules):
 - Encode expert heuristics (examples):
 - IF humidity is low AND wind is high $\rightarrow$ increase wind coefficient (C_U) .
 - IF elevation > threshold $\rightarrow$ apply elevation correction (C_M) .
 - Rules can choose which formula variant to use or set/adjust coefficients based on observed conditions.

4. **Calculation layer:** use formulas (Christiansen or sub-equations) to compute ET using the final coefficients decided by rules.
5. **Inference & explanation:** RADEX's inference engine explains results (shows which rules fired and which data produced coefficients).
6. **Output:** ET estimates, diagnostic messages, "why/how" explanation, and possibly recommended irrigation actions.

Advantages to mention in exam:

- Combines **expert knowledge** (rules) with **precise computations** (formulas).
- Can handle conditional logic (special cases) that pure formula alone cannot.
- Provides **explanations** ("why" an adjustment was made).

Limitations to mention:

- Quality depends on correctness of rules and input data.
 - May need calibration for local conditions.
 - Complexity if many rules interact (conflict resolution needed).
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